

NLP-Powered Job Skill Intelligence & Demand Analytics

PROJECT REPORT

Natural Language Processing • Machine Learning • Skill Intelligence

JOB SKILL EXTRACTION FROM JOB DESCRIPTIONS USING NLP

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1. INTRODUCTION

1.1 Project Background

The recruitment industry generates a large volume of job descriptions containing valuable information about required technical skills, tools, technologies, qualifications, experience, and job responsibilities. Manually analyzing these descriptions to identify frequently demanded skills is time-consuming, inconsistent, and difficult to scale.

Natural Language Processing (NLP) provides techniques for automatically processing and understanding human language. In this project, NLP techniques are applied to job descriptions to automatically identify and extract technical and professional skills.

The project, “**Job Skill Extraction from Job Descriptions using NLP**,” focuses on building an automated pipeline capable of processing large-scale job descriptions, extracting relevant skills, normalizing different skill representations, categorizing skills, and analyzing skill demand across job roles.

The project also incorporates statistical NLP, machine learning, semantic techniques, Named Entity Recognition (NER), and transformer-based approaches to improve extraction quality.

1.2 Natural Language Processing Overview

NLP is a branch of Artificial Intelligence that enables computers to process, analyze, and understand human language.

Major NLP techniques used or explored in this project include:

- Text cleaning
- Tokenization
- Stop-word removal
- Stemming
- Lemmatization
- N-gram analysis
- TF-IDF
- Word embeddings
- Semantic similarity
- Named Entity Recognition
- Machine learning classification
- Transformer-based language models

1.3 Job Skill Extraction

Job skill extraction is the process of automatically identifying skills mentioned in job descriptions.

For example:

Input:

“The candidate should have experience in Python, SQL, Power BI, Excel and data visualization.”

Extracted Skills:

- Python
- SQL
- Power BI
- Excel
- Data Visualization

The extracted information can subsequently be used for recruitment analytics, job matching, candidate ranking, skill-gap analysis, and workforce planning.

2. PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Organizations receive and publish thousands of job descriptions containing unstructured textual information. Identifying the technical skills required for different positions manually is inefficient and difficult to scale.

There is therefore a need for an automated NLP-based solution that can:

1. Process large volumes of job descriptions.
2. Identify relevant skills automatically.
3. Handle different skill names and variations.
4. Normalize extracted skills.
5. Categorize skills.
6. Analyze skill demand across job roles.
7. Present insights through an interactive dashboard.

2.2 Business Objective

The primary business objective is to transform unstructured job descriptions into structured skill intelligence that can support recruitment and workforce-related decisions.

The solution can help organizations understand:

- Which skills are most demanded.
- Which skills are associated with specific job roles.
- Which technology categories are growing in demand.
- What skills recruiters commonly mention.
- How skill requirements vary across roles.

2.3 Technical Objective

The technical objectives are:

- Develop an NLP preprocessing pipeline.
- Create a skill taxonomy.
- Implement rule-based extraction.
- Explore statistical NLP techniques.
- Implement NER-based extraction.
- Develop an ML-based classification pipeline.
- Explore transformer-based extraction.
- Normalize extracted skills.
- Evaluate extraction performance.
- Develop a visualization/dashboard layer.
- Provide an application/API interface for skill extraction.

2.4 Expected Outcome

The expected outcome is an end-to-end **Job Skill Intelligence System** that converts raw job descriptions into structured skill information and provides analytical insights through visualization.

3. PROJECT SCOPE & USE CASE

3.1 Project Scope

The project covers the following stages:

Job Dataset → Data Understanding → Cleaning → NLP Preprocessing → Skill Taxonomy → Skill Extraction → Normalization → Categorization → Evaluation → Dashboard → Application/API

3.2 Recruitment Use Case

Recruiters can enter or upload a job description and receive a list of identified skills.

For example:

Job Description	Extracted Skills
Data Analyst	SQL, Python, Power BI, Excel
Data Scientist	Python, Machine Learning, SQL, TensorFlow
Frontend Developer	React, JavaScript, HTML, CSS
Cloud Engineer	AWS, Docker, Kubernetes

3.3 Practical Applications

The developed system can support:

- Automated job description analysis
- Recruitment analytics
- Skill demand analysis

- Candidate-job matching
- Resume screening
- Skill-gap analysis
- Candidate ranking
- Workforce planning
- Training recommendation systems

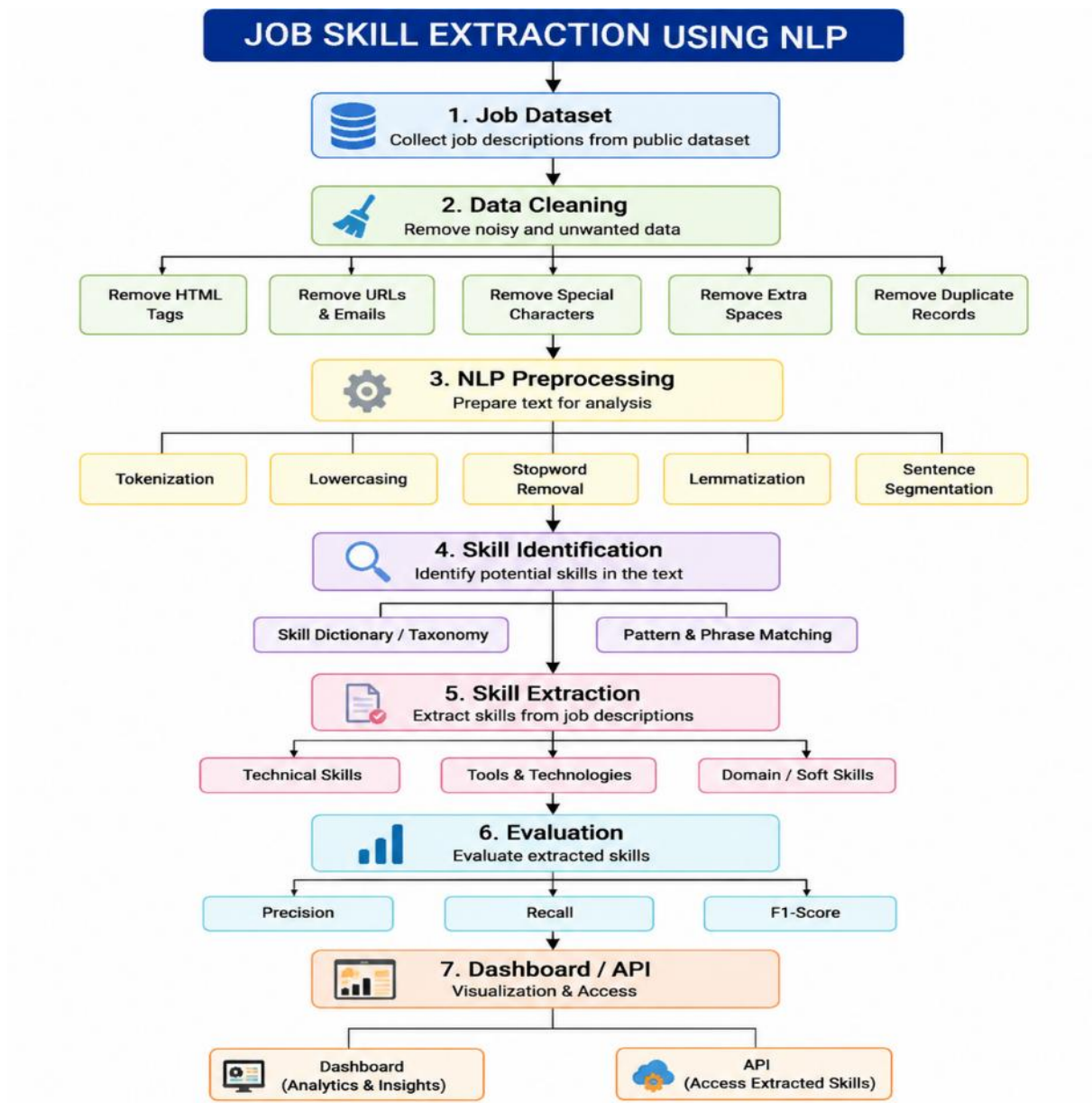
4. TECHNOLOGY & TOOLS USED

Technology/Tool	Purpose
Python	Core programming language
Pandas	Dataset processing
NumPy	Numerical operations
NLTK	NLP preprocessing
spaCy	NLP and NER
Scikit-learn	Machine learning
TF-IDF	Text feature extraction
Transformers	Transformer-based NLP
BERT/DistilBERT/RoBERTa	Advanced language modeling
Matplotlib	Visualization
Seaborn	Statistical visualization
Power BI	Business intelligence dashboard
Streamlit	Application interface
FastAPI	API development
Jupyter Notebook	Development and experimentation
VS Code	Application development
Git	Version control

5. SYSTEM ARCHITECTURE & METHODOLOGY

5.1 System Architecture

The proposed system follows a multi-stage architecture:



5.2 Methodology

The methodology consists of:

1. Dataset collection
2. Data understanding
3. Exploratory Data Analysis
4. Data cleaning
5. NLP preprocessing
6. Skill taxonomy development
7. Rule-based extraction
8. Statistical and semantic analysis
9. NER
10. Machine learning
11. Transformer-based extraction
12. Model evaluation
13. Error analysis
14. Skill normalization

15. Dashboard development
16. Application/API development

6. DATASET COLLECTION & UNDERSTANDING

6.1 Dataset

The project uses a large job-description dataset containing approximately:

1,615,940 job records

The cleaned dataset contains **19 relevant columns**.

6.2 Important Dataset Fields

Field	Description
Job Id	Unique job identifier
Experience	Required experience
Qualifications	Educational qualifications
Salary Range	Salary information
Location	Job location
Country	Country of employment
Benefits	Benefits offered with the job
skills	Skills required or mentioned for the job
Work Type	Full-time/part-time/remote etc.
Company Size	Company classification
Job Posting Date	Date of posting
Preference	Candidate preference
Contact Person	Recruiter/contact information
Contact	Contact details
Job Title	Job position
Role	Job role/category
Job Portal	Source of job posting
Job Description	Detailed job requirements
Company	Additional information or description about the company

6.3 Data Inspection

Initial inspection was performed using Pandas to identify:

- Number of records
- Number of columns
- Data types

- Missing values
- Duplicate records
- Unique job titles
- Text length
- Frequently occurring terms

7. EXPLORATORY DATA ANALYSIS

EDA was performed to understand the characteristics and distribution of the dataset.

Key EDA Areas

- Job title distribution
- Role distribution
- Location distribution
- Work type distribution
- Experience distribution
- Qualification distribution
- Salary range
- Job posting trends
- Job description length
- Missing values
- Duplicate records

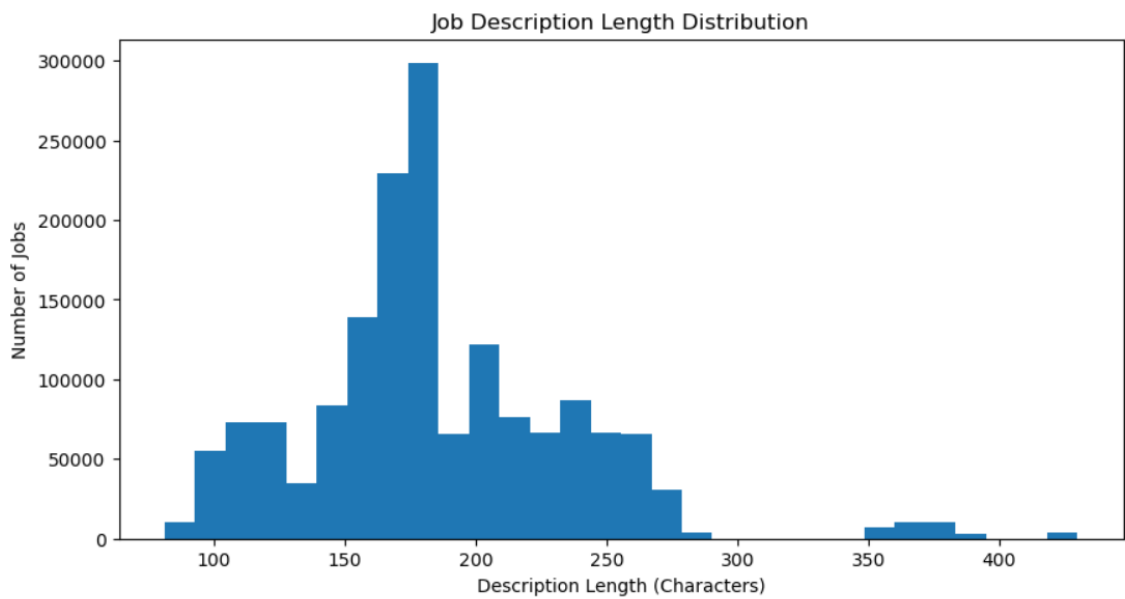
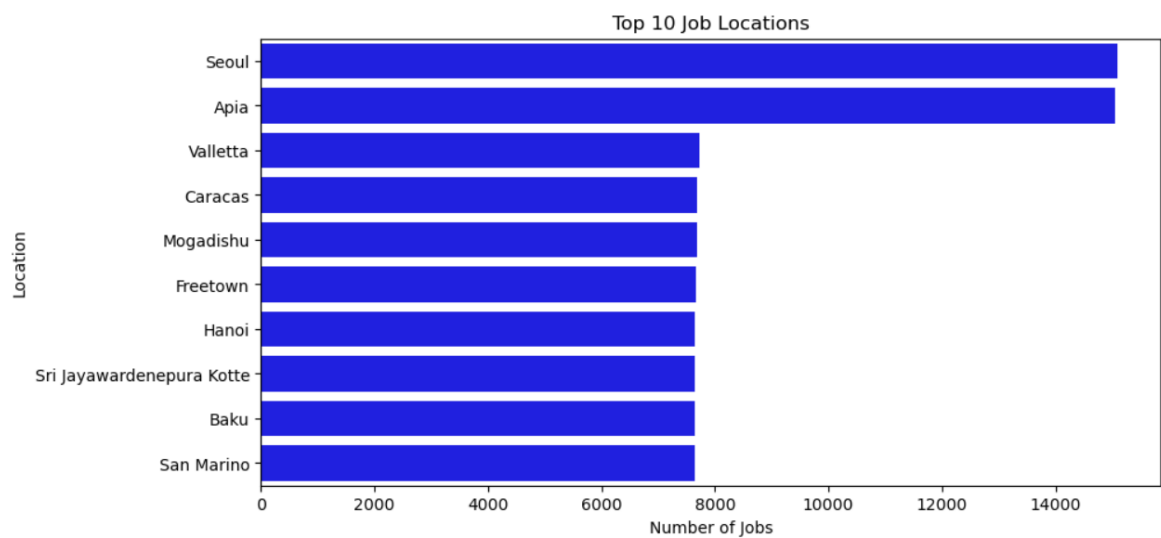
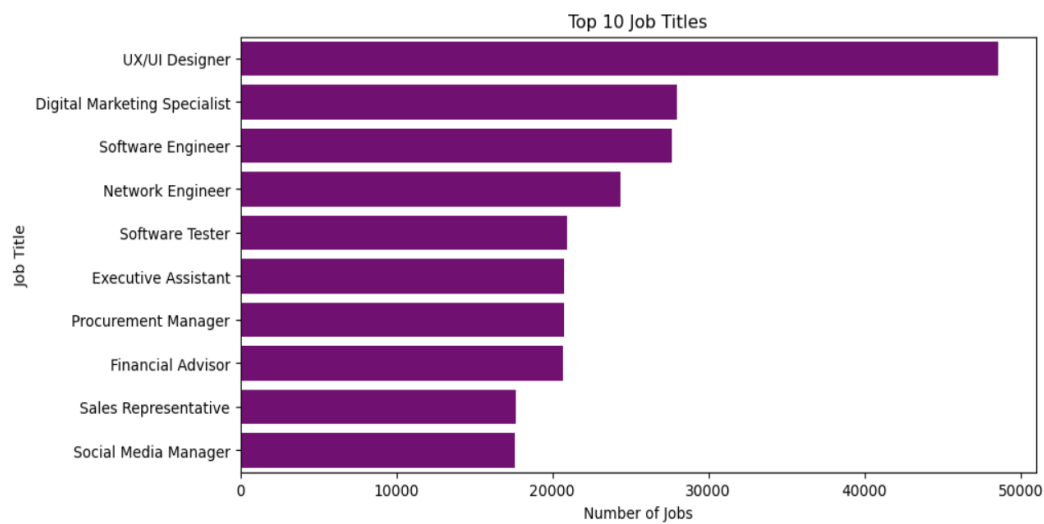
Example Questions

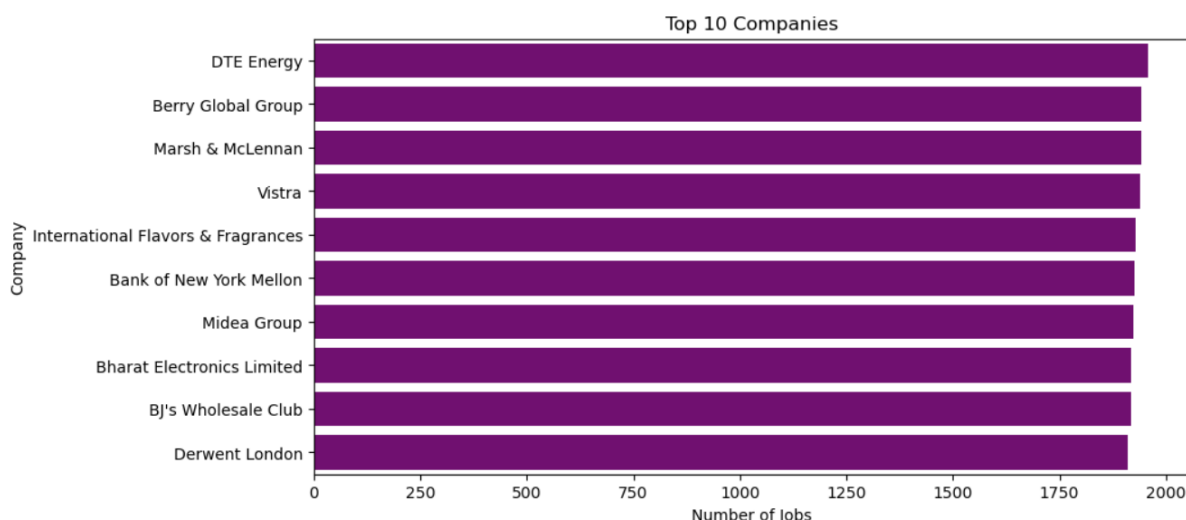
EDA was designed to answer questions such as:

- What are the most common job roles?
- Which locations contain the highest number of jobs?
- What is the average job description length?
- Which job categories dominate the dataset?
- How many records contain missing descriptions?
- Which job roles have the highest skill requirements?

Visualization Examples

- Top Job Roles
- Jobs by Location
- Job Description Length Distribution
- Jobs by Work Type
- Experience Distribution
- Job Posting Trend
- Missing Value Analysis





8. DATA CLEANING & NLP PREPROCESSING

Raw job descriptions frequently contain HTML tags, URLs, email addresses, special characters, duplicated spaces, and irrelevant textual elements.

8.1 Data Cleaning

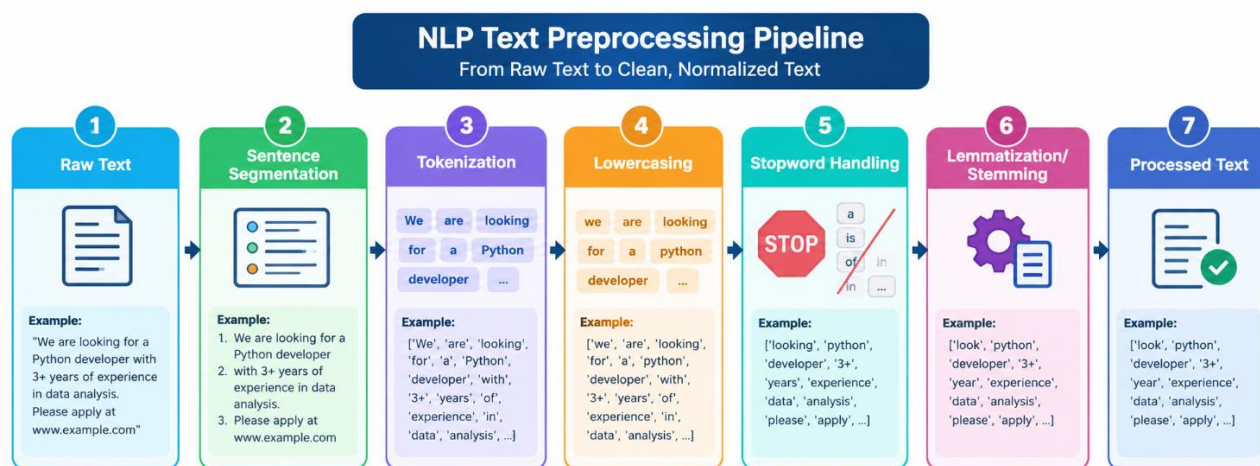
The cleaning process includes:

- Removing HTML tags
- Removing URLs
- Removing email addresses
- Removing phone numbers
- Removing unnecessary special characters
- Removing extra spaces
- Handling missing descriptions
- Removing duplicate text where applicable

8.2 NLP Preprocessing

After cleaning, NLP preprocessing was applied to prepare the text for further analysis.

The preprocessing pipeline includes:



8.3 Tokenization

Tokenization divides a job description into smaller units called tokens.

For example:

Input:

"Experience in Python and SQL is required."

Tokens:

Python | SQL | Experience | required

8.4 Stopword Handling

Common words that provide limited information for certain NLP tasks can be removed. Examples include:

- the
- is
- and
- of
- in
- for
- with

However, stopwords removal was applied carefully because some short words can be meaningful in technical terminology.

8.5 Stemming

Stemming reduces words to their root form by removing word endings.

For example:

- developing → develop
- developed → develop

- development → develop

8.6 Lemmatization

Lemmatization converts words to their meaningful base form using linguistic information.

For example:

- running → run
- studies → study
- technologies → technology

Lemmatization generally produces more meaningful results than simple stemming and was therefore considered useful for semantic text processing.

8.7 Sentence Segmentation

Job descriptions were divided into sentences to identify skill mentions within meaningful textual contexts such as:

- Required Skills
- Technical Skills
- Responsibilities
- Qualifications
- Preferred Skills

```
JOB TITLE:
Digital Marketing Specialist

1. ORIGINAL:
Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.

2. TOKENS:
['Social', 'Media', 'Managers', 'oversee', 'an', 'organizations', 'social', 'media', 'presence', '.', 'They', 'create', 'and', 'schedule', 'content', 'engage', 'with', 'followers', ',', 'and', 'analyze', 'social', 'media', 'metrics', 'to', 'drive', 'brand', 'awareness', 'and', 'engagement', '.']

3. LOWERCASE:
social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.

4. WITHOUT STOPWORDS:
['Social', 'Media', 'Managers', 'oversee', 'organizations', 'social', 'media', 'presence', 'create', 'schedule', 'content', 'engage', 'followers', 'analyze', 'social', 'media', 'metrics', 'drive', 'brand', 'awareness', 'engagement']

5. LEMMATIZATION:
['Social', 'Media', 'Managers', 'oversee', 'organization', 'social', 'medium', 'presence', 'create', 'schedule', 'content', 'engage', 'follower', 'analyze', 'social', 'medium', 'metric', 'drive', 'brand', 'awareness', 'engagement']

6. STEMMING:
['social', 'media', 'manag', 'overse', 'organ', 'social', 'media', 'presenc', 'creat', 'schedul', 'content', 'engag', 'follow', 'analyz', 'social', 'media', 'metric', 'drive', 'brand', 'awar', 'engag']

7. SENTENCES:
['Social Media Managers oversee an organizations social media presence.', 'They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.']

8. FINAL CLEAN TEXT:
Social Media Managers oversee organization social medium presence create schedule content engage follower analyze social medium metric drive brand awareness engagement
```

9. SKILL TAXONOMY DEVELOPMENT

A skill taxonomy was developed to provide a structured representation of technical and professional skills.

9.1 Skill Dictionary

The dictionary contains commonly used technologies, programming languages, databases, cloud platforms, frameworks, analytical tools, and other job-related skills.

Examples include:

Programming Languages

- Python
- Java
- JavaScript
- C++
- R

Databases

- MySQL
- PostgreSQL
- Oracle
- MongoDB
- SQL Server

Data & Analytics

- SQL
- Power BI
- Tableau
- Excel
- Pandas
- NumPy
- Statistics

Machine Learning

- Machine Learning
- Deep Learning
- TensorFlow
- PyTorch
- Scikit-learn

Cloud & DevOps

- AWS
- Azure
- Google Cloud
- Docker
- Kubernetes

Web Technologies

- HTML

- CSS
- React
- Angular
- Node.js

9.2 Skill Categories

Skills were grouped into logical categories to support dashboard analysis.

Category	Examples
Programming	Python, Java, C++, R
Database	SQL, MySQL, MongoDB
Data Analytics	Power BI, Tableau, Excel
Machine Learning	Scikit-learn, TensorFlow
Cloud	AWS, Azure, GCP
Web Development	React, Angular, Node.js
DevOps	Docker, Kubernetes, Jenkins
Data Engineering	Spark, Hadoop, ETL
Business Skills	Communication, Leadership
Statistical	Statistics, Probability

9.3 Aliases and Synonyms

Different job descriptions may refer to the same technology using different names.

Examples:

Alias	Canonical Skill
PowerBI	Power BI
MS SQL	SQL Server
Postgre SQL	PostgreSQL
JS	JavaScript
ReactJS	React
ML	Machine Learning
NLP	Natural Language Processing

This mapping improves consistency in the final skill analysis.

	skill	category	aliases
0	Python	Programming	python, python3
1	Java	Programming	java
2	C++	Programming	c++, cpp
3	JavaScript	Programming	javascript, js
4	HTML	Web Development	html, html5
5	CSS	Web Development	css, css3
6	React	Web Development	react, reactjs, react.js
7	Angular	Web Development	angular, angularjs
8	SQL	Database	sql
9	MySQL	Database	mysql
10	PostgreSQL	Database	postgres, postgresql
11	MongoDB	Database	mongodb, mongo
12	Power BI	Data Analytics	power bi, powerbi, power-bi
13	Tableau	Data Analytics	tableau
14	Excel	Data Analytics	excel, ms excel, microsoft excel
15	Pandas	Data Analytics	pandas
16	NumPy	Data Analytics	numpy
17	Machine Learning	Machine Learning	machine learning, ml

Total skills in taxonomy: 51

Categories:

category

Data Analytics 5

Programming 4

Web Development 4

Database 4

Machine Learning 4

DevOps 4

Marketing 4

Soft Skills 4

Cloud 3

Big Data 3

Networking 3

Project Management 3

Human Resources 3

Engineering 3

Name: count, dtype: int64

10. RULE-BASED SKILL EXTRACTION

Rule-based extraction was implemented as one of the fundamental skill extraction approaches.

10.1 Dictionary Matching

The job description was compared against the predefined skill dictionary.

If a skill or recognized variation was found, it was added to the extracted skill list.

Example

Input:

"The candidate should have strong experience in Python, SQL, Power BI and Excel."

Output:

```
Python
SQL
Power BI
Excel
```

10.2 Regular Expressions

Regular expressions were used to identify specific patterns and variations.

For example, patterns can identify:

- SQL
- SQL Server
- Power BI
- C++
- C#
- .NET
- Node.js

Regex improves the ability to detect skills that may not match simple word boundaries.

10.3 Phrase Matching

spaCy PhraseMatcher can be used to detect multi-word skills such as:

- Machine Learning
- Deep Learning
- Natural Language Processing
- Data Visualization
- Power BI
- Microsoft Azure
- Google Cloud Platform

10.4 Advantages

The rule-based approach provides:

- Fast execution
- Easy implementation
- High interpretability
- Easy modification
- No requirement for large labeled datasets

10.5 Limitations

The major limitations include:

- Dependence on the skill dictionary
- Difficulty handling unseen skills
- Difficulty understanding context
- Potential false positives
- Difficulty recognizing new terminology

	Job Title	Job Description	extracted_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations...	[Social Media, Content Creation, Content Sched...
1	Web Developer	Frontend Web Developers design and implement u...	[Web Development]
2	Operations Manager	Quality Control Managers establish and enforce...	[Quality Control, Quality Assurance]
3	Network Engineer	Wireless Network Engineers design, implement, ...	[Wireless Networking, Troubleshooting]
4	Event Manager	A Conference Manager coordinates and manages c...	[Conference Management, Budget Management, Ven...
5	Software Tester	A Quality Assurance Analyst tests software and...	[Quality Control, Quality Assurance, Bug Track...
6	Teacher	A Classroom Teacher educates students in a spe...	[Teaching, Lesson Planning, Student Assessment]
7	UX/UI Designer	User Interface Designers focus on the visual a...	[Web Development, UI Design, UX/UI Designer]

11. STATISTICAL & SEMANTIC NLP TECHNIQUES

Multiple NLP techniques were explored to understand important terms and relationships within job descriptions.

11.1 TF-IDF

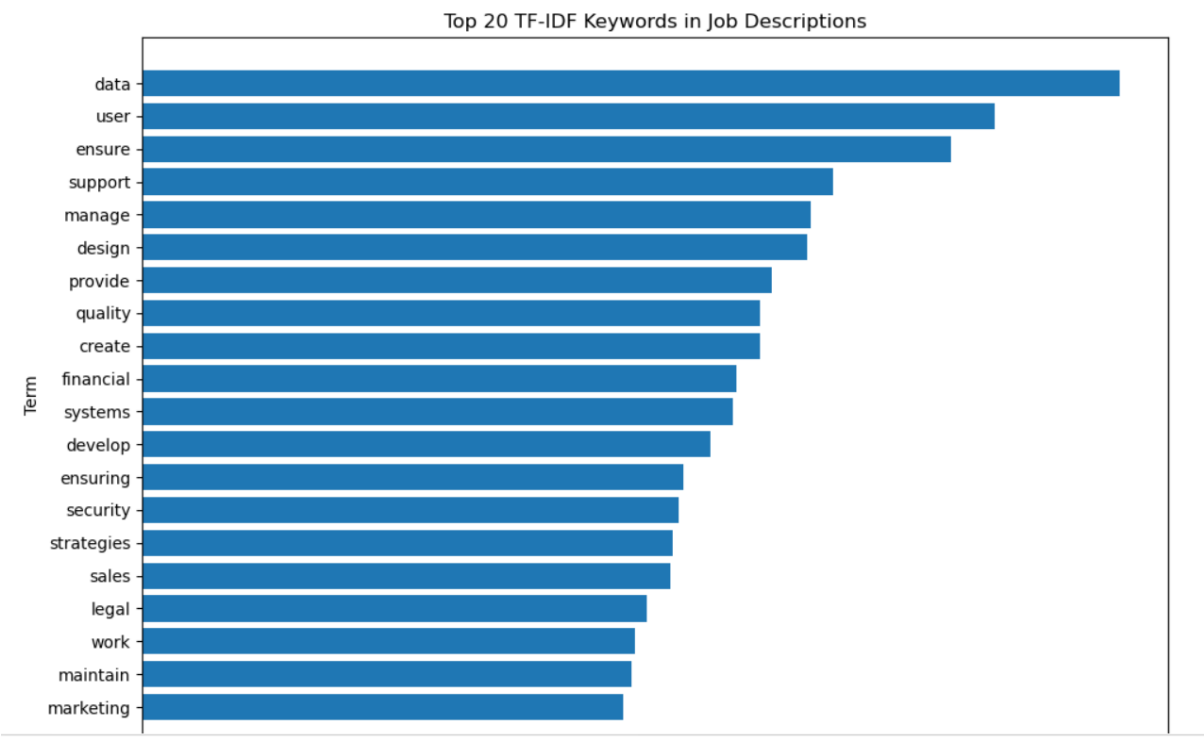
Term Frequency–Inverse Document Frequency measures the importance of a term within a document relative to the complete collection of documents.

TF-IDF helps identify terms that are important in particular job descriptions or job categories.

For example, terms such as:

- Python
- SQL
- AWS
- React
- Power BI

may receive high importance depending on the job category.



	Job Title	Job Description	skills	regex_skills	phrase_skills	extracted_skills	tfidf_terms
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[social media, media, social, awareness engagement, followers, followers analyze, organizations social, media presence, media metrics, content engage]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]	[developers, web, user, developers design, interfaces websites, frontend web, collaborate designers, frontend, user interfaces, appealing user]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[quality, quality control, control, maintain product, service quality, service, develop quality, organization develop, perform, quality standards]

11.2 N-Gram Analysis

N-grams identify sequences of words.

Unigrams

Single words:

- Python
- SQL
- Analytics

Bigrams

Two-word combinations:

- Machine Learning
- Data Science
- Power BI
- Deep Learning

Trigrams

Three-word combinations:

- Natural Language Processing
- Data Visualization Tools
- Machine Learning Model

N-gram analysis is particularly useful for identifying multi-word technical skills.

	Job Title	Job Description	ngram_text
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	frontend web developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. they collaborate with designers and backend developers to create seamless web experiences for users.
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control managers establish and enforce quality standards within an organization. they develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	wireless network engineers design, implement, and maintain wireless network solutions. they optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	a conference manager coordinates and manages conferences, meetings, and events. they plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	a quality assurance analyst tests software and products to ensure they meet quality standards. they identify defects, report issues, and work with development teams to resolve problems.
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	a classroom teacher educates students in a specific subject or grade level. they create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive	user interface designers focus on the visual and interactive aspects of digital interfaces. they design layouts, buttons, and other elements to ensure a cohesive

```
display(unigram_df.head(20))
```

	term	score
0	data	0.034448
1	user	0.032906
2	ensure	0.027128
3	support	0.023379
4	manage	0.021880
5	quality	0.020908
6	provide	0.020214
7	design	0.020136
8	create	0.019818
9	systems	0.019468
10	security	0.018889
11	financial	0.018655
12	sales	0.018301
13	ensuring	0.017513
14	customer	0.017211
15	develop	0.017183

```
display(bigram_df.head(20))
```

	term	score
0	social media	0.013160
1	decision making	0.010179
2	managers oversee	0.009771
3	supply chain	0.009348
4	visually appealing	0.009189
5	provide legal	0.008534
6	mergers acquisitions	0.008160
7	designers create	0.008136
8	provide insights	0.008071
9	ensure compliance	0.007936
10	products services	0.007813
11	digital interfaces	0.007610
12	strategies manage	0.007484
13	user experience	0.007417
14	diagnose treat	0.007016
15	user interface	0.006992
16	data driven	0.006907
17	quality standards	0.006847

display(trigram_df.head(20))			display(ngram_df.head(20))		
	term	score		ngram	tfidf_score
0	illnesses perform surgeries	0.008072	0	data	0.025550
1	implement security measures	0.007425	1	user	0.021088
2	speech language disorders	0.007000	2	ensure	0.019991
3	strong client relationships	0.007000	3	support	0.017205
4	design implement maintain	0.006743	4	design	0.016846
5	drive revenue growth	0.006687	5	manage	0.016637
6	provide legal support	0.006000	6	quality	0.016373
7	designers specialize designing	0.005860	7	financial	0.016316
8	analyze financial data	0.005661	8	provide	0.016114
9	decision making organization	0.005587	9	create	0.015265
10	cross functional teams	0.005509	10	systems	0.015239
11	provide administrative support	0.005499	11	develop	0.014771
12	conduct user research	0.005367			
13	data provide insights	0.005344			
14	legal matters related	0.005280			

11.3 Word Embeddings

Word embeddings represent words as numerical vectors.

Techniques considered include:

- Word2Vec
- GloVe
- Sentence Transformers

Embedding-based methods can identify relationships between words based on their contextual or semantic similarity.

11.4 Semantic Similarity

Semantic similarity can identify terms that are conceptually related even when the exact wording is different.

For example:

"Data visualization platform"

may have semantic similarity with:

"Power BI"

This approach can improve extraction beyond exact keyword matching.

	Job Title	semantic_skills
0	Digital Marketing Specialist	[[Social Media Management, 0.7816]]
1	Web Developer	[[Frontend Development, 0.7237], [Backend Development, 0.666], [Web Development, 0.6003]]
2	Operations Manager	[[Quality Management Systems, 0.7435], [Quality Control Processes, 0.7408], [Quality Control, 0.7061], [Quality Assurance, 0.5843], [Quality Standards, 0.5607]]
3	Network Engineer	[[Wireless Networking, 0.5725], [Wireless Security, 0.5585], [Wireless Network Design, 0.5494]]
4	Event Manager	[[Conference Management, 0.7453], [Conference Logistics, 0.577], [Event Management, 0.5487]]
5	Software Tester	[[Quality Assurance, 0.6964], [Software Testing, 0.5004]]
6	Teacher	[[Classroom Management, 0.5853], [Teaching, 0.5734], [Lesson Planning, 0.563]]
7	UX/UI Designer	[[User Interface, 0.6639], [UI Design, 0.6187], [Visual Design, 0.5796], [UX Design, 0.5192], [Interaction Design, 0.5029]]
8	UX/UI Designer	[[Interaction Design, 0.6797], [Micro-interaction Design, 0.56], [User Interface, 0.5333], [User Experience, 0.5144], [UX Design, 0.5001]]
9	Wedding Planner	[]
10	QA Analyst	[[Performance Testing, 0.7475], [Software Testing, 0.6154], [Performance Monitoring, 0.6072], [Load Testing, 0.5807], [Performance Analysis, 0.5598]]
11	Litigation Attorney	[[Family Law, 0.6299]]
12	Mechanical Engineer	[[Mechanical Engineering, 0.6858], [Mechanical Design, 0.6687], [Mechanical Systems, 0.6194], [Mechanical Components, 0.546]]
13	Network Administrator	[[Network Security, 0.6514], [Intrusion Prevention, 0.6009], [Firewall Administration, 0.5345], [Security Policies, 0.5338]]
14	Account Manager	[[Account Management, 0.6176]]
15	Brand Manager	[[Product Branding, 0.6677], [Product Marketing, 0.5906], [Brand Identity, 0.5751]]

	Job Title	regex_skills	phrase_skills	semantic_skill_names	final_extracted_skills
0	Digital Marketing Specialist	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media Management]	[Social Media, Social Media Analytics, Content Creation, Community Engagement, Social Media Management]
1	Web Developer	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Backend Development, Web Development]
2	Operations Manager	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]	[Quality Management Systems, Quality Control Processes, Quality Control, Quality Assurance, Quality Standards]	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection, Quality Management Systems, Quality Assurance]
3	Network Engineer	[Wireless Networking]	[Wireless Networking]	[Wireless Networking, Wireless Security, Wireless Network Design]	[Wireless Networking, Wireless Security, Wireless Network Design]
4	Event Manager	[Event Management, Conference Management, Event Logistics, Vendor Coordination]	[Conference Management, Event Management, Event Planning, Conference Logistics, Budget Management, Vendor Coordination]	[Conference Management, Conference Logistics, Event Management]	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]
5	Software Tester	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]	[Quality Assurance, Quality Standards]	[Quality Assurance, Software Testing]	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]
6	Teacher	[Teaching, Curriculum Development, Lesson Planning, Instruction]	[Teaching, Lesson Planning, Instruction]	[Classroom Management, Teaching, Lesson Planning]	[Teaching, Curriculum Development, Lesson Planning, Instruction, Classroom Management]
7	UX/UI Designer	[User Interface, UI Design]	[User Interface, User Interaction]	[User Interface, UI Design, Visual Design, UX Design, Interaction Design]	[User Interface, UI Design, User Interaction, Visual Design, UX Design, Interaction Design]

12. NAMED ENTITY RECOGNITION (NER)

Named Entity Recognition is an NLP technique used to identify meaningful entities in text.

12.1 Standard NER

Standard NLP models can identify entities such as:

- Organizations
- Locations
- Persons
- Dates
- Products

However, standard NER models are generally not specifically trained to identify technical skills.

12.2 Custom Skill Entities

A custom NER approach can define domain-specific entities such as:

- SKILL
- TECHNOLOGY
- TOOL
- FRAMEWORK
- PROGRAMMING_LANGUAGE
- DATABASE
- CLOUD_PLATFORM

Example

Input:

"The candidate must have experience in Python, SQL, AWS and Power BI."

Entities:

Entity	Type
Python	SKILL
SQL	DATABASE/SKILL
AWS	CLOUD_PLATFORM
Power BI	TOOL

12.3 Custom NER Approach

A custom NER pipeline can be developed using:

1. Annotated job descriptions
2. Entity labels
3. spaCy training pipeline
4. Training and validation datasets
5. Model evaluation
6. Error analysis

Custom NER provides a more context-aware approach than simple dictionary matching.

	Job Title	standard_ner
0	Digital Marketing Specialist	[(Social Media Managers, ORG)]
1	Web Developer	[]
2	Operations Manager	[(Quality Control Managers, ORG)]
3	Network Engineer	[(Wireless Network Engineers, ORG)]
4	Event Manager	[]
5	Software Tester	[(Quality Assurance, ORG)]
6	Teacher	[]
7	UX/UI Designer	[(Interface Designers, ORG)]
8	UX/UI Designer	[(Interaction Designers, ORG)]
9	Wedding Planner	[]
10	QA Analyst	[]
11	Litigation Attorney	[]
12	Mechanical Engineer	[(Mechanical Design Engineers, ORG)]
13	Network Administrator	[]
14	Account Manager	[]
15	Brand Manager	[]
16	Social Worker	[]

	Job Title	Job Description	custom_ner
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[(Social Media, SKILL)]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	[]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	[(Quality Control, SKILL), (quality standards, SKILL), (quality control processes, SKILL)]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	[]
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	[]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	[(Quality Assurance, SKILL), (quality standards, SKILL)]
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	[(instruction, SKILL)]

13. MACHINE LEARNING MODEL

Machine learning was explored to automatically learn patterns associated with skills and job-related terminology.

13.1 Training Data

The training dataset can contain text examples labeled according to whether a term or phrase represents a skill.

Example:

Text	Label
Python	SKILL
SQL	SKILL
salary	NON-SKILL
location	NON-SKILL
Power BI	SKILL

13.2 Feature Engineering

TF-IDF was used as a text feature representation technique.

The text is transformed into numerical vectors that can be provided to machine learning algorithms.

13.3 ML Classification Pipeline

The general pipeline is:

Job Description → Text Preprocessing → TF-IDF Vectorization → ML Classifier → Skill Prediction

13.4 Candidate Models

Scikit-learn models that can be considered include:

- Logistic Regression
- Naive Bayes
- Linear SVM

Logistic Regression provides a simple and interpretable baseline for text classification.

13.5 Advantages

- Learns patterns from training data
- Can generalize beyond manually defined rules
- Supports automated classification
- Easy to integrate with existing NLP pipelines

13.6 Limitations

- Requires labeled training data

- Performance depends on data quality
- May struggle with unseen terminology
- Context understanding is limited compared with transformer models

Training dataset shape: (5485, 3)

	Text	Entity	Label
0	brand awareness	brand awareness	DIGITAL_MARKETING
1	social media	social media	DIGITAL_MARKETING
2	frontend	frontend	WEB_DEVELOPMENT
3	backend	backend	WEB_DEVELOPMENT
4	quality control	quality control	QUALITY_CONTROL
5	wireless network	wireless network	NETWORKING
6	network	network	NETWORKING
7	logistics	logistics	EVENT_MANAGEMENT
8	quality assurance	quality assurance	TESTING
9	instruction	instruction	EDUCATION
10	user interface	user interface	UI_UX
11	budget management	budget management	MANAGEMENT
12	performance testing	performance testing	TESTING
13	legal guidance	legal guidance	LEGAL
14	legal matters	legal matters	LEGAL
15	family law	family law	LEGAL

Description training dataset: (4373, 2)

	Text	Label
0	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	DIGITAL_MARKETING
1	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	WEB_DEVELOPMENT
2	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	QUALITY_CONTROL
3	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	NETWORKING
4	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	EVENT_MANAGEMENT
5	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	TESTING
6	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	EDUCATION
7	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	UI_UX
8	A Wedding Consultant assists couples in planning and coordinating their weddings, helping them with vendor selection, budget management, and ensuring a seamless wedding day experience.	MANAGEMENT
9	Performance Testing Specialists assess the performance of software applications by conducting load, stress, and scalability tests. They identify bottlenecks, optimize performance, and ensure software can handle user demands effectively.	TESTING

14. TRANSFORMER-BASED MODEL

Transformer-based NLP models provide advanced contextual understanding of language.

Models considered in the project include:

- BERT
- DistilBERT
- RoBERTa

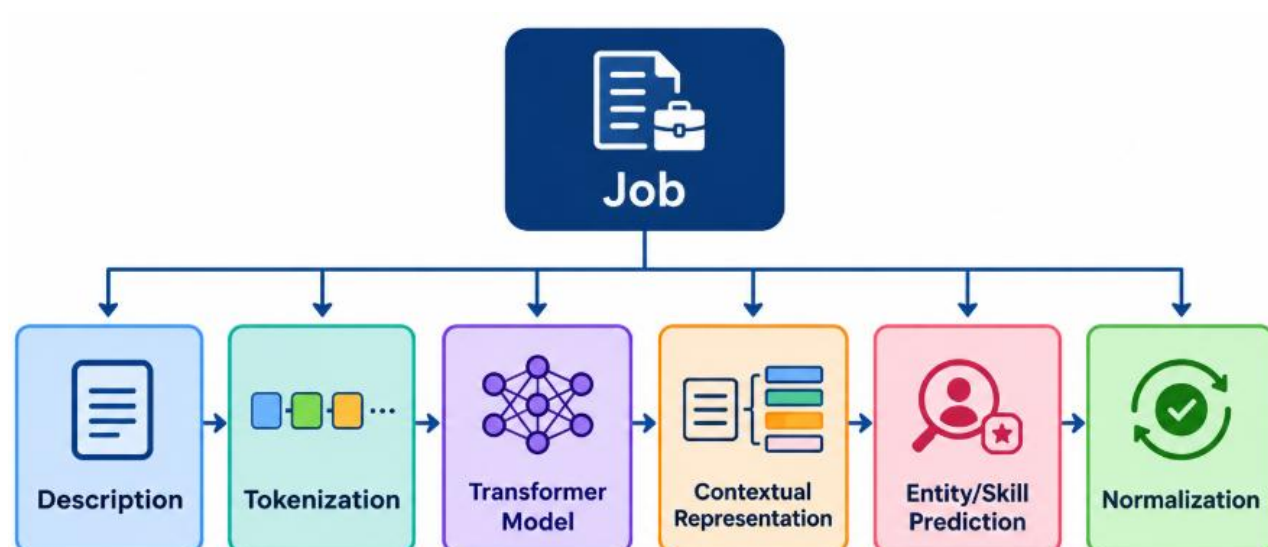
14.1 BERT

BERT uses bidirectional contextual representations to understand the relationship between words within a sentence.

For example, the word "Java" can be interpreted differently depending on its surrounding context.

14.2 Transformer Extraction Approach

The extraction process can be represented as:



14.3 Advantages

Transformer models provide:

- Better contextual understanding
- Improved recognition of variations
- Better handling of complex language
- Ability to identify context-dependent entities

14.4 Limitations

Transformer models generally require:

- More computational resources
- Training or fine-tuning data
- Longer processing time
- More complex deployment

For this reason, transformer-based approaches are considered complementary to faster rule-based methods.

DAY 14 TRANSFORMER RESULTS

=====

Precision : 0.00%
 Recall : 0.00%
 F1 Score : 0.00%
 Accuracy : 93.33%

Confusion Matrix:

	Predicted O	Predicted B-SKILL	Predicted I-SKILL
Actual O	126	0	0
Actual B-SKILL	5	0	0
Actual I-SKILL	4	0	0

Classification Report:

	precision	recall	f1-score	support
O	0.93	1.00	0.97	126
B-SKILL	0.00	0.00	0.00	5
I-SKILL	0.00	0.00	0.00	4
accuracy			0.93	135
macro avg	0.31	0.33	0.32	135
weighted avg	0.87	0.93	0.90	135

Transformer:
DistilBERT

Precision:
0.00%

Recall:
0.00%

F1 Score:
0.00%

Accuracy:
93.33%

15. MODEL EVALUATION & COMPARISON

Model evaluation is important for determining how accurately the system identifies relevant skills.

15.1 Evaluation Metrics

Precision

Precision measures how many extracted skills are actually correct.

Precision = True Positives / (True Positives + False Positives)

Recall

Recall measures how many actual skills were successfully identified.

Recall = True Positives / (True Positives + False Negatives)

F1-Score

F1-score combines precision and recall.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Accuracy

Accuracy measures the percentage of correct predictions among all predictions.

15.2 Confusion Matrix

A confusion matrix can be used to analyze:

- True Positive
- True Negative
- False Positive
- False Negative

15.3 Method Comparison

Method	Speed	Context Understanding	Dictionary Required	Generalization
Dictionary	High	Low	Yes	Low
Regex	High	Low	Partially	Medium
PhraseMatcher	High	Low-Medium	Yes	Medium
TF-IDF	High	Medium	No	Medium
NER	Medium	High	No/Training dependent	High
ML	Medium	Medium	No	Medium-High
Transformer	Low-Medium	Very High	No	High

The final system can combine multiple approaches to achieve a balance between speed, interpretability, and extraction quality.

	Job Title	Actual Skills	Dictionary	Regex	TF-IDF	NER	ML Model	Transformer
0	Digital Marketing Specialist	[social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising]	☐	[social media]	☐	☐	☐	☐
1	Web Developer	[html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)]	☐	☐	☐	☐	☐	☐
2	Operations Manager	[quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge]	☐	☐	☐	☐	☐	☐
3	Network Engineer	[wireless network design and architecture wi-fi standards and protocols rf (radio frequency) planning and optimization wireless security protocols troubleshooting wireless network issues]	☐	☐	☐	☐	☐	☐
4	Event Manager	[event planning conference logistics budget management vendor coordination marketing and promotion client relations]	☐	☐	☐	☐	☐	☐
5	Software Tester	[quality assurance processes testing methodologies (e.g., manual, automated) bug tracking and reporting test case development regression testing]	☐	☐	☐	☐	☐	☐
6	Teacher	[teaching pedagogy classroom management curriculum development student assessment differentiated instruction]	☐	☐	☐	☐	☐	☐
7	UX/UI Designer	[ui design principles and best practices graphic design tools (e.g., adobe photoshop, illustrator) typography and color theory visual design and layout responsive design]	☐	[user interface]	☐	☐	☐	☐

DAY 15 MODEL COMPARISON

	Model	TP	FP	FN	Precision	Recall	F1 Score	Accuracy
0	Dictionary	12	34	3412	26.09	0.35	0.69	0.0
1	Regex	4	271	3420	1.45	0.12	0.22	0.0
2	TF-IDF	8	11	3416	42.11	0.23	0.46	0.0
3	NER	4	26	3420	13.33	0.12	0.23	0.0
4	ML Model	12	34	3412	26.09	0.35	0.69	0.0
5	Transformer	0	0	3424	0.00	0.00	0.00	0.0

=====

BEST MODEL

=====

Best model based on F1: Dictionary

Precision: 26.09%

Recall: 0.35%

F1 Score: 0.69%

Accuracy: 0.00%

Dictionary

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1964	36

Regex

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1771	229

TF-IDF

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1981	19

NER

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1976	24

16. ERROR ANALYSIS

Error analysis was performed to identify cases where the extraction system may produce incorrect or incomplete results.

16.1 False Positives

A false positive occurs when a non-skill term is incorrectly identified as a skill.

For example, a common word may have the same spelling as a technology name.

16.2 False Negatives

A false negative occurs when a valid skill mentioned in a job description is not detected.

Possible causes include:

- Skill missing from dictionary
- Unusual terminology
- Abbreviations
- Spelling variations
- Contextual expressions

16.3 Partial Entities

Some skills may appear as part of a larger phrase.

For example:

"Microsoft Power BI Desktop"

may require extraction of the canonical skill:

Power BI

16.4 Synonyms and Variations

The same technology may be written differently:

- ReactJS
- React.js
- React JS

These need to be mapped to the same canonical skill.

16.5 Spelling Issues

Job descriptions may contain:

- Typographical errors
- Inconsistent capitalization
- Missing punctuation
- Different separators

Normalization and fuzzy/semantic matching can help reduce these problems.

=====

ERROR ANALYSIS RESULT

=====

Total error records: 13063

	Model	Text	Actual	Predicted	Error	Error Detail
0	Dictionary	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising		False Negative	social media platforms (e.g., facebook, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising, twitter
1	Dictionary	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)		False Negative	react, css, angular) user experience (ux), javascript frontend frameworks (e.g., html
2	Dictionary	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge		False Negative	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge

=====

ERROR CATEGORY SUMMARY

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	Model	Error	Count
0	Dictionary	False Negative	2000
2	Dictionary	Partial Entity	34
1	Dictionary	False Positive	28
4	Dictionary	Wrong Entity	28
3	Dictionary	Spelling Problem	7
5	ML Model	False Negative	2000
7	ML Model	Partial Entity	34
6	ML Model	False Positive	28
9	ML Model	Wrong Entity	28
8	ML Model	Spelling Problem	7
10	NER	False Negative	2000
12	NER	Partial Entity	30
11	NER	False Positive	24

=====

MODEL-WISE ERROR COUNT

=====

	Model	Total Errors
3	Regex	2764
0	Dictionary	2097
1	ML Model	2097
2	NER	2081
4	TF-IDF	2024
5	Transformer	2000

False Positive

	Model	Actual	Predicted	Error Detail
98	Dictionary	sql (structured query language) database design query optimization data modeling database maintenance problem-solving skills	sql	sql
106	Dictionary	creative writing copywriting advertising copy content creation brand storytelling marketing campaigns proofreading and editing	writing	writing
124	Dictionary	backend development restful apis database integration java ee (enterprise edition) performance optimization version control (e.g., git)	java	java

False Negative

	Model	Actual	Predicted	Error Detail
0	Dictionary	social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising	social media platforms (e.g., facebook, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising, twitter	
1	Dictionary	html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)	react, css, angular) user experience (ux), javascript frontend frameworks (e.g., html	
2	Dictionary	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge	

Partial Entity

	Model	Actual	Predicted	Error Detail
100	Dictionary	sql (structured query language) database design query optimization data modeling database maintenance problem-solving skills	sql	Predicted entity partially matches actual
108	Dictionary	creative writing copywriting advertising copy content creation brand storytelling marketing campaigns proofreading and editing	writing	Predicted entity partially matches actual
126	Dictionary	backend development restful apis database integration java ee (enterprise edition) performance optimization version control (e.g.,	java	Predicted entity partially matches actual

Unknown Skill

	Model	Actual	Predicted	Error Detail
2112	Regex	ui design principles and best practices graphic design tools (e.g., adobe photoshop, illustrator) typography and color theory visual design and layout responsive design	user interface	Predicted skill not present in actual annotation
2140	Regex	front-end web development html, css, javascript responsive design web performance optimization cross-browser compatibility	user experience	Predicted skill not present in actual annotation
2141	Regex	front-end web development html, css, javascript responsive design web performance optimization cross-browser compatibility	user interface	Predicted skill not present in actual annotation

Spelling Problem

	Model		Actual	Predicted	Error Detail
236	Dictionary	java programming java frameworks (e.g.	java	java	Possible spelling variation
526	Dictionary	azure) devops practices automation security in the cloud disaster recovery scalability	azure	azure	Possible spelling variation
757	Dictionary	java programming java frameworks (e.g.	java	java	Possible spelling variation

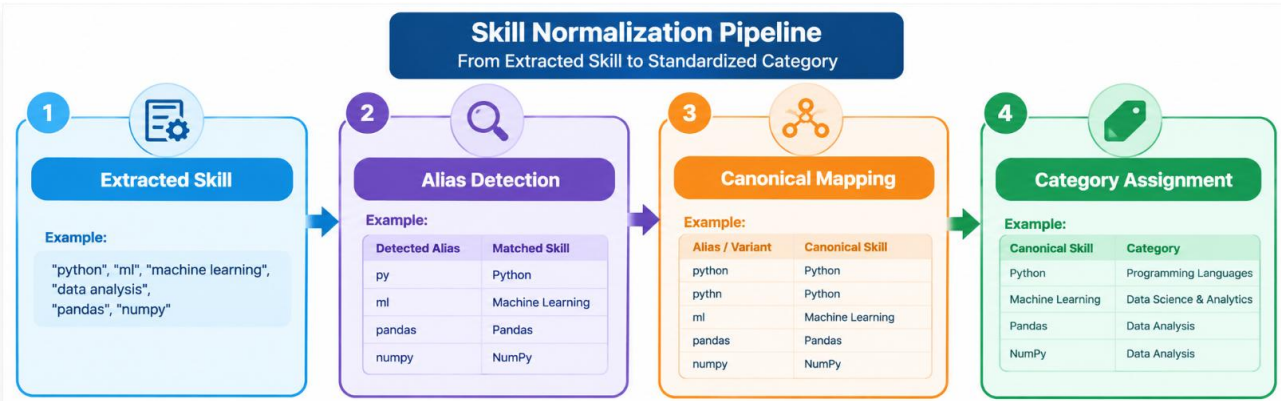
17. SKILL NORMALIZATION & CATEGORIZATION

Skill normalization converts multiple representations of the same skill into a single standard form.

Example

Extracted Form	Normalized Skill	Category
Python 3	Python	Programming
Python Programming	Python	Programming
ReactJS	React	Web Development
React.js	React	Web Development
PowerBI	Power BI	Data Analytics
MS SQL	SQL Server	Database
ML	Machine Learning	AI/ML

17.1 Normalization Process



17.2 Benefits

Normalization provides:

- Consistent reporting
- Accurate skill frequency
- Better dashboard results
- Improved job matching

- Easier comparison between job roles

Job Title		skills	Normalized Skills
0	Digital Marketing Specialist	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	Social Media Platforms, Facebook, Twitter, Instagram Content Creation And Scheduling Social Media Analytics And Insights Community Engagement Paid Social Advertising
1	Web Developer	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	HTML, CSS, Javascript Frontend Frameworks, React, Angular User Experience Ux
2	Operations Manager	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	Quality Control Processes And Methodologies Statistical Process Control Spc Root Cause Analysis And Corrective Action Quality Management Systems, Iso 9001 Compliance And Regulatory Knowledge
3	Network Engineer	Wireless network design and architecture Wi-Fi standards and protocols RF (Radio Frequency) planning and optimization Wireless security protocols Troubleshooting wireless network issues	Wireless Network Design And Architecture Wi Fi Standards And Protocols Rf Radio Frequency Planning And Optimization Wireless Security Protocols Troubleshooting Wireless Network Issues
4	Event Manager	Event planning Conference logistics Budget management Vendor coordination Marketing and promotion Client relations	Event Planning Conference Logistics Budget Management Vendor Coordination Marketing And Promotion Client Relations
5	Software Tester	Quality assurance processes Testing methodologies (e.g., manual, automated) Bug tracking and reporting Test case development Regression testing	Quality Assurance Processes Testing Methodologies, Manual, Automated Bug Tracking And Reporting Test Case Development Regression Testing
6	Teacher	Teaching pedagogy Classroom management Curriculum development Student assessment Differentiated instruction	Teaching Pedagogy Classroom Management Curriculum Development Student Assessment Differentiated Instruction
7	UX/UI Designer	UI design principles and best practices Graphic design tools (e.g., Adobe Photoshop, Illustrator) Typography and color theory Visual design and layout Responsive design	Ui Design Principles And Best Practices Graphic Design Tools, Adobe Photoshop, Illustrator Typography And Color Theory Visual Design And Layout Responsive Design
8	UX/UI Designer	Interaction design principles User behavior and psychology Wireframing and prototyping tools Animation and micro-interaction design Collaborative design processes	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes
9	Wedding Planner	Wedding planning Vendor coordination Event management	Wedding Planning Vendor Coordination Event Management

ACTUAL RAW SKILLS → CANONICAL SKILLS

	Raw Skill	Cleaned Skill	Canonical Skill	Frequency
0	Python	python	Python	98
1	Google Analytics	google analytics	Google Analytics	86
2	Java	java	Java	80
3	CSS	css	CSS	74
4	Sketch	sketch	Sketch	73
5	AWS	aws	AWS	64
6	Interaction design principles User behavior and psychology Wireframing and prototyping tools Animation and micro-interaction design Collaborative design processes	interaction design principles user behavior and psychology wireframing and prototyping tools animation and micro interaction design collaborative design processes	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes	61
7	Tableau	tableau	Tableau	58
8	Illustrator) Typography and color theory Visual design and layout Responsive design	illustrator typography and color theory visual design and layout responsive design	Illustrator Typography And Color Theory Visual Design And Layout Responsive Design	53
9	Adobe Photoshop	adobe photoshop	Adobe Photoshop	53
10	UI design principles and best practices Graphic design tools (e.g.	ui design principles and best practices graphic design tools	Ui Design Principles And Best Practices Graphic Design Tools	53
11	Network management Troubleshooting Network security IT certifications (e.g.	network management troubleshooting network security it certifications	Network Management Troubleshooting Network Security It Certifications	52
12	CCNA)	ccna	Ccna	52
13	Node.js	nodejs	Node.js	49

TOP NORMALIZED SKILLS

=====

	Canonical Skill	Frequency
0	Python	98
1	Google Analytics	86
2	Java	80
3	CSS	74
4	Sketch	73
5	Tableau	68
6	AWS	64
7	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes	61
8	Adobe Photoshop	53
9	Illustrator Typography And Color Theory Visual Design And Layout Responsive Design	53
10	Ui Design Principles And Best Practices Graphic Design Tools	53
11	Ccna	52
12	Network Management Troubleshooting Network Security It Certifications	52
13	Media	40

18. DASHBOARD & DATA VISUALIZATION

A **Job Skill Intelligence Dashboard** was designed to transform extracted skill data into meaningful business insights.

18.1 Dashboard KPIs

Key performance indicators include:

- Total Jobs
- Unique Skills
- Top Skill
- Top Job Role
- Most Required Technology
- Top Skill Category

18.2 Dashboard Visualizations

Recommended dashboard visuals include:

Top Skills

A bar chart displaying the most frequently demanded skills.

Skill Demand by Job Role

Shows which skills are most commonly associated with different job roles.

Skill Category Distribution

Displays the distribution of:

- Programming
- Database
- Cloud
- Analytics
- Machine Learning
- Web Development
- DevOps

Skill Trends

Shows changes in skill demand over job posting dates where sufficient date information is available.

Job Role Analysis

Provides comparison of skill requirements across different job categories.

18.3 Business Value

The dashboard converts unstructured job descriptions into an interactive analytical solution that helps stakeholders identify technology and skill demand.



19. APPLICATION / API DEVELOPMENT

An application layer can be developed to allow users to interact with the skill extraction system.

19.1 Application Interface

A Streamlit-based interface can provide:

- Job description input
- Text upload
- Extract Skills button
- Extracted skill results
- Skill categories
- Normalized skills
- Downloadable results

19.2 Input

Example input:

"We are looking for a Data Analyst with experience in Python, SQL, Power BI, Excel and Tableau."

19.3 Output

Extracted Skills:

1. **Python**
 2. **SQL**
 3. **Power BI**
 4. **Excel**
 5. **Tableau**
-

Skill Category:

Programming: Python

Database: SQL

Analytics: Power BI, Excel, Tableau

19.4 Extraction Pipeline

```
graph TD
    A[User Input] --> B[Text Cleaning]
    B --> C[NLP Preprocessing]
    C --> D[Skill Detection]
    D --> E[Skill Normalization]
    E --> F[Category Mapping]
    F --> G[Result Generation]
```

19.5 API Architecture

A FastAPI implementation can expose an endpoint such as:

POST /extract-skills

The API receives a job description and returns structured skill information in JSON format.

Example conceptual response:

```
{
  "skills": [
    "Python",
    "SQL",
    "Power BI",
    "Excel"
  ]
}
```

20. TESTING & RESULTS

Testing was conducted to verify that the extraction pipeline works correctly under different input conditions.

20.1 Test Cases

Test Case	Input	Expected Result
TC01	Python, SQL	Python, SQL extracted
TC02	Power BI, Tableau	Both skills detected
TC03	ReactJS	React normalized
TC04	ML	Machine Learning normalized
TC05	Empty description	Appropriate validation message
TC06	Duplicate skills	Duplicate output removed
TC07	Mixed case text	Skills still detected
TC08	Multi-word skills	Correct phrase extraction

20.2 Sample Result

Input

"Required skills include Python, SQL, Power BI, Tableau, Excel and Machine Learning."

Output

Extracted Skills

- Python
- SQL

- Power BI
- Tableau
- Excel
- Machine Learning

Categories

- Programming
- Database
- Data Analytics
- Machine Learning

20.3 Final Results

The project successfully demonstrates that NLP can be used to convert large-scale unstructured job descriptions into structured skill information.

The combination of dictionary matching, phrase matching, statistical NLP, NER, machine learning, semantic techniques, and transformer-based approaches provides multiple levels of skill extraction.

21. CHALLENGES & LIMITATIONS

21.1 Challenges

Several challenges were encountered during development:

Large Dataset

Processing approximately **1.6 million job records** requires efficient memory and processing strategies.

Unstructured Text

Job descriptions have inconsistent formatting and writing styles.

Skill Variations

The same skill can be represented using multiple names.

Ambiguous Terms

Some words may represent either a general term or a technical skill depending on context.

New Technologies

Technology terminology continuously evolves, making static dictionaries difficult to maintain.

Lack of Labeled Data

Supervised ML and custom NER approaches require accurately labeled training data.

Computational Requirements

Transformer models require significantly more computational resources than rule-based techniques.

21.2 Limitations

The current system may have limitations related to:

- Unseen skills
 - Misspelled technologies
 - Ambiguous terminology
 - Incomplete job descriptions
 - Limited labeled training data
 - Context-dependent skill meanings
 - Dataset bias
 - Variation between job portals
-

22. CONCLUSION

The **Job Skill Extraction from Job Descriptions using NLP** project successfully demonstrates the application of Natural Language Processing, machine learning, semantic analysis, and visualization techniques to solve a practical recruitment analytics problem.

The system processes job descriptions, cleans and preprocesses textual information, identifies relevant skills, normalizes different skill representations, categorizes skills, and generates structured skill information.

Multiple approaches including **dictionary matching, regular expressions, phrase matching, TF-IDF, N-gram analysis, embeddings, NER, machine learning, and transformer-based models** were explored.

The extracted information can be further analyzed through a **Job Skill Intelligence Dashboard**, providing insights into skill demand, job roles, technology categories, and workforce requirements.

Overall, the project provides a foundation for building intelligent recruitment solutions that can reduce manual effort and support data-driven hiring decisions.
